

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Common To All Branches

Semester :II

Subject Code & Name: BTBS202 Engineering Chemistry

Max Marks: 60

Date: 20/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Water which forms lather with soap is -----	1	1
	a. Soft water b. Hard water c. Neutral water d. Acidic water		
2	Temporary hardness can be removed by -----	1	1
	a. EDTA method b. Mohr's method c. Boiling d. Precipitation		
3	If the water is neutral then it having the pH-----	1	1
	a. Zero b. 1-7 c. 7 d. 7-14		
4	A system consisting of Oxygen and Nitrogen gases is -----	1	1
	a. One phase system b. Two phase system c. Three phase system d. None of these		
5	For the water system no. of component is -----	1	1
	a. 3 b. 1 c. 2 d. None of these		
6	At ----- the corrosion is greater	2	1
	a. More basic b. More acidic c. Neutral d. a & c both		
7	When a buried pipeline is protected from corrosion by connecting to Mg block. It is called-----	2	1
	a. Electrolysis b. Extraction of metal c. Impressed voltage protection d. Scarification of metal		

8	Galvanizing is the process of coating -----over the protective metal.				2	1
	a. Sn	b. Zn	c. Cu	d. Ni		
9	Fuel is a combustible substance, containing ----- as a main constituent.				1	1
	a. Hydrogen	b. Oxygen	c. Nitrogen	d. Carbon		
10	Grease is an example of -----				1	1
	a. Solid lubricant	b. Liquid lubricant	c. Semi-solid lubricant	d. None of these		
11	The conductivity of an electrolyte is due to the-----				1	1
	a. Presence of impurity in the electrolyte	b. Free movement of ions in the solution	c. Reunion of ions in the solution	d. Release of heat energy due to ionization		
12	Phenolphthalein has pH range is-----				1	1
	a. 8.2-10.5	b. 4.3 - 5.4	c. 4.4 - 5.0	d. 4.1 - 5.3		
Q. 2	Solve the following.					12
A)	Explain Zeolite process of softening of water with its advantages and disadvantages.				2	6
B)	How does the Hardness of water determined by EDTA Complexometric method?				1	6
Q.3	Solve the following.					12
A)	Explain Gibbs Phase Rule with 1) Phase 2) No. of Component terminologies.				1	6
B)	Explain the Phase diagram of one component Water system.				2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain how corrosion of metals controlled by sacrificial anode method.				2	6
B)	Explain factors influencing rate of corrosion.				1	6
C)	Explain Hydrogen evolution mechanism of wet corrosion.				2	6

Q.5	Solve Any Two of the following.		12
A)	Define Fuel. Explain any five characteristics of good fuel in detail.	1	6
B)	Explain how the percentage of carbon and Hydrogen in coal can be estimated.	2	6
C)	Describe in brief solid and liquid lubricants.	2	6
Q. 6	Solve Any Two of the following.		12
A)	Explain conductometric titration with suitable example.	2	6
B)	Write note on: Glass electrode.	1	6
C)	Explain Ostwald's theory of acid – base indicator.	2	6
	*** End ***		